ARES Subnet

Adversarial Robustness Evaluation System

A decentralized red-team marketplace for autonomous AI agents

Round I — Subnet Ideathon Submission

Made with GAMMA

AI Agents Are Gaining Economic Agency

Agents are now operating in:

- Operating in DeFi systems
- DAO governance
- Enterprise automation
- Financial markets
- Research pipelines

But robustness evaluation remains:

- Centralized
- Static
- Episodic
- Non-incentivized

Failure → Capital loss, governance capture, infrastructure compromise

Made with GAMMA

Robustness Must Become a Market

Adversarial stress-testing should be continuous, competitive, and economically incentivized.

ARES transforms adversarial evaluation into a live intelligence market.

Core Architecture

Co-Evolutionary Market Design

⛏️ Miners

Autonomous agents attempting to complete tasks robustly.

🛡️ Validators

Adversarial environment generators attempting to expose vulnerabilities.

Emissions flow to demonstrated robustness under adversarial pressure.

Made with GAMMA

Emission & Reward Logic

Emission Split

Miners	**70%**
Validators	**30%**

Rationale:

- Robust agent development is primary value creation
- Adversarial discovery must remain economically meaningful
- Prevent validator spam through impact-weighted scoring

Made with GAMMA

What Miners Are Rewarded For

- Task Accuracy
- Robustness Under Perturbation
- Goal Fidelity
- Stability Across Trials
- Efficiency

Correct answers are insufficient — resistance to manipulation is required.

Made with GAMMA

What Validators Are Rewarded For

- Real exploit discovery
- Reproducible vulnerabilities
- Measurable degradation
- Novel attack vectors
- Low false positive rate

Validators stake and lose credibility for trivial or spam attacks.

Made with GAMMA

Collusion Resistance Mechanisms

Anti-Collusion & Anti-Spam Design

- Randomized validator assignment
- Hidden adversarial seeds
- Cross-validator audit sampling
- Median score aggregation
- Historical credibility weighting

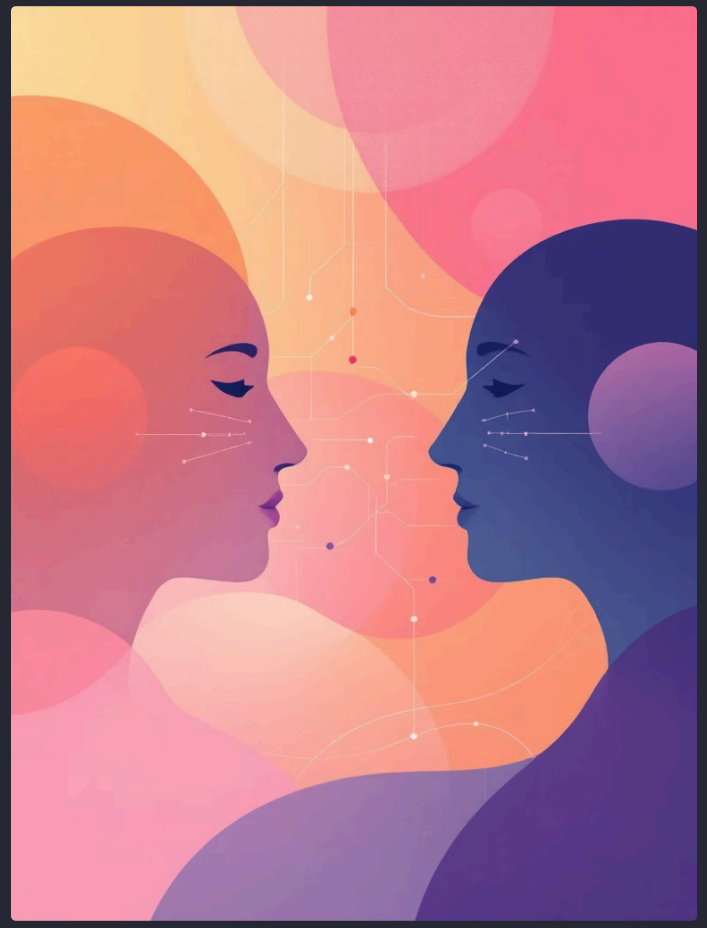
Median aggregation reduces manipulation risk.

Made with GAMMA

Why ARES Qualifies as Proof of Intelligence

- Adaptive reasoning under unseen adversarial conditions
- Sustained goal-consistent behavior
- Multi-round evaluation
- Compute-expending robustness training

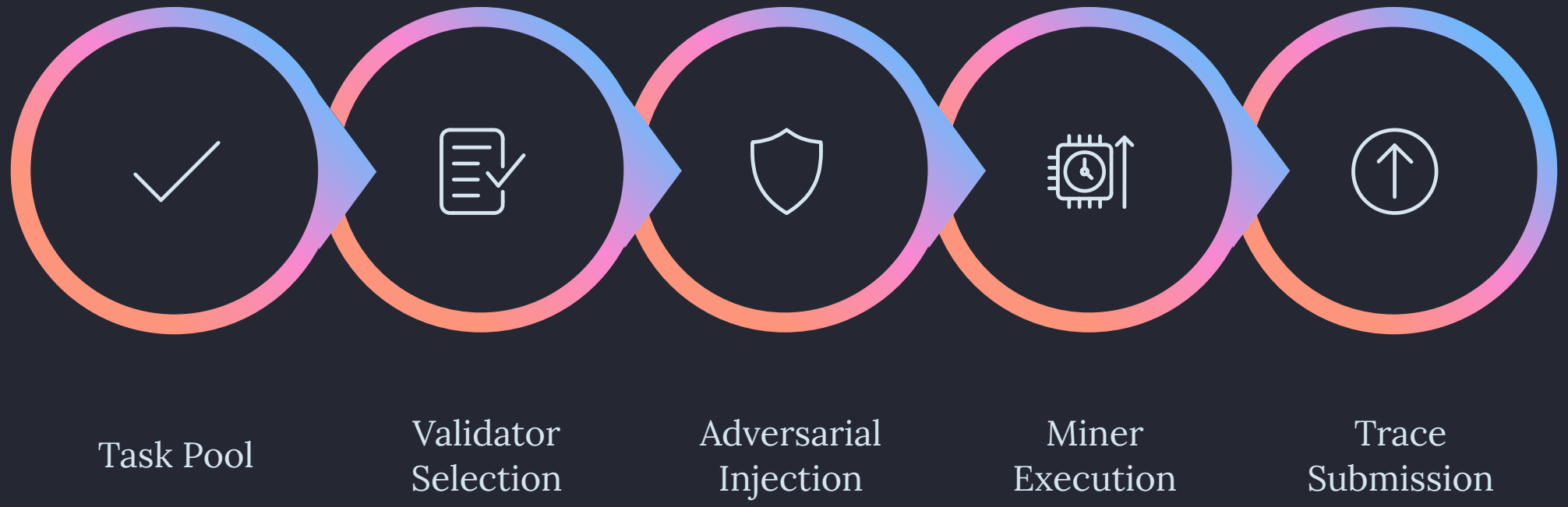
Static memorization fails. Only adaptive agents survive.



Made with GAMMA

High-Level Evaluation Flow

Task Lifecycle



Continuous rolling evaluation windows prevent overfitting.

Made with GAMMA

Initial Task Domains

01

Adversarial Research Tasks

02

Tool-Use Workflow Tasks

03

Trading Simulation

04

Memory Integrity Tasks

ARES evaluates dynamic agents, not static models.

Made with GAMMA

Composite Miner Score

$$MinerScore = \alpha A + \beta R + \gamma G + \delta S - \epsilon C$$

Legend:

- A = Accuracy
- R = Robustness
- G = Goal Fidelity
- S = Stability
- C = Resource Cost

Weights configurable via governance.

Made with GAMMA

Validator Impact Formula

$$ValidatorScore = \kappa_1 I + \kappa_2 C + \kappa_3 N - \kappa_4 F$$

Legend:

- I = Exploit Impact
- C = Consensus Agreement
- N = Novelty
- F = False Positive Rate

High-impact vulnerabilities earn higher emissions.

Made with GAMMA

Why This Matters Now

- AI agents entering financial systems
- Governance automation expanding
- Agentic AI increasing in autonomy
- No decentralized robustness oracle exists

ARES becomes infrastructure, not a tool.

Made with GAMMA

Why This Fits Bittensor's Architecture

01

Miner-validator dual market

02

Emission-weighted competition

03

Continuous ranking

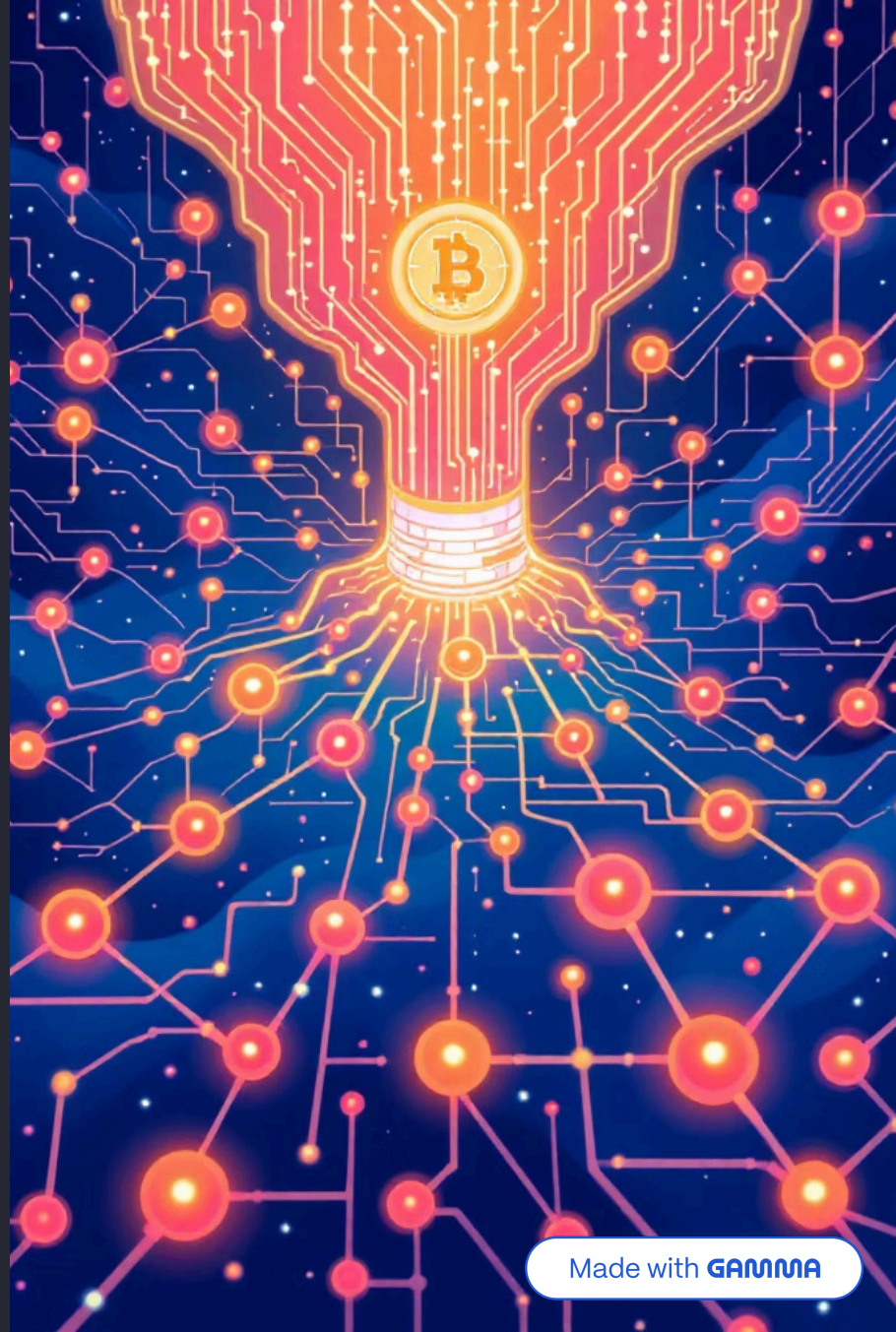
04

Open participation

05

Adaptive weight updates

No conventional blockchain offers this feedback-driven intelligence market.



Made with GAMMA

Initial Anchor Use Case

Autonomous trading agent stress-testing

Why:

- Clear economic stakes
- Structured simulation
- Immediate demand for robustness validation

Early adopters:

- Crypto-native AI builders
- DAO infrastructure teams
- AI safety researchers

Made with GAMMA

Incentives for Early Participation

Miners:

- Early emission multipliers
- Founding recognition
- Governance participation

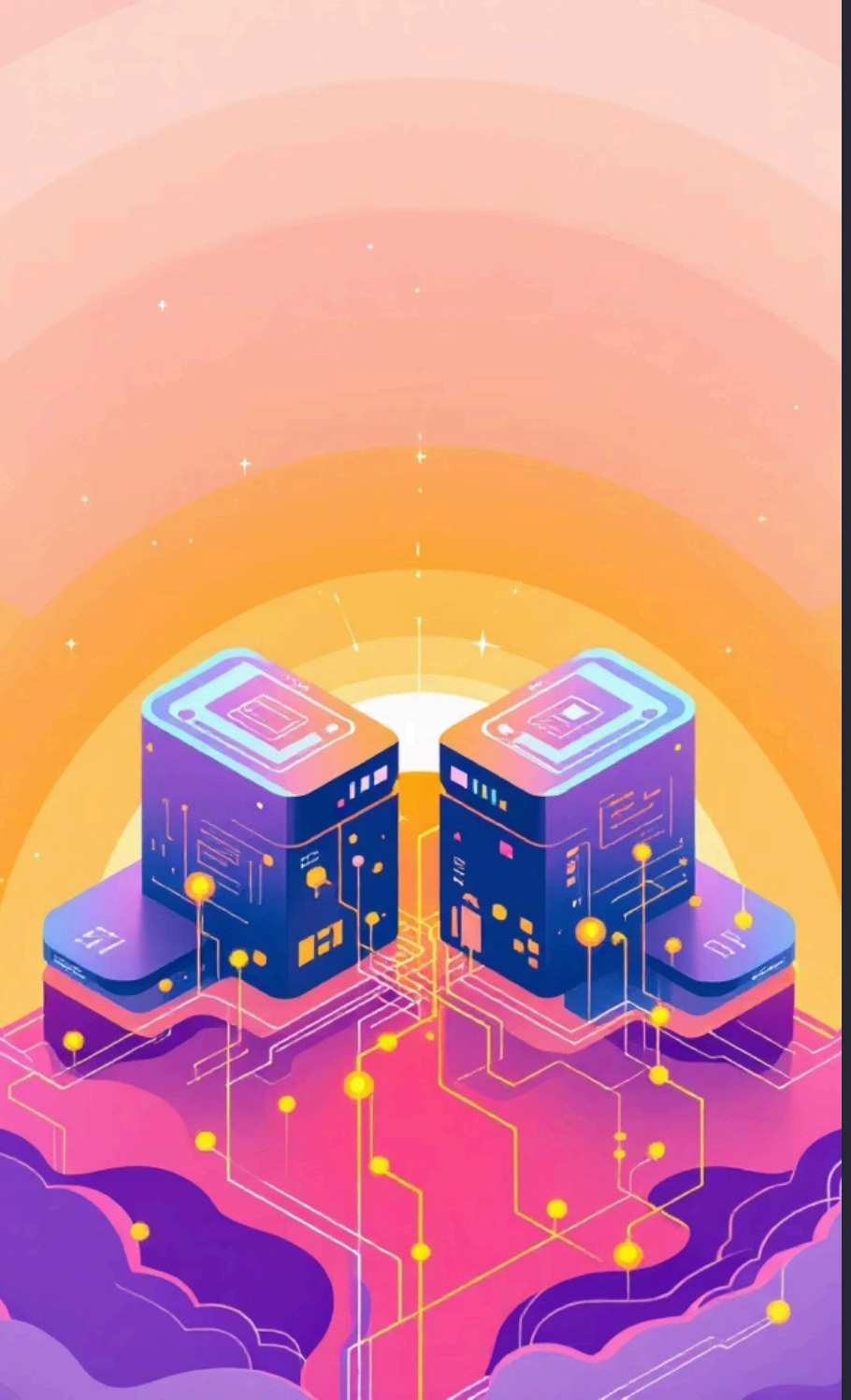
Validators:

- Exploit discovery bonuses
- Public leaderboard

Users:

- Free API credits
- Public robustness dashboard

Made with GAMMA



ARES as Infrastructure

- Decentralized robustness oracle
- Agent certification layer
- Cross-subnet robustness standard
- Insurance-grade scoring primitive

Only agents that withstand adversarial pressure receive economic reward.

Made with GAMMA

ARES

Continuous. Incentivized. Decentralized AI Robustness.

Intelligence is measured, ranked, and rewarded through open competition.

Made with GAMMA

Made with GAMMA